KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

School of Computer Engineering | SAP BTP Developer Project

Record-to-Report (R2R) — Month-End & Year-End Financial Close

SAP FI/CO Module + SAP BTP Developer Extension

Organisation	**Specialization**	**Academic Year**	**Topic**
Vinayak Associate Lucknow, U.P, 226001.	SAP BTP Developer FI/CO + BTP Extension	2025-2026	Record-to-Report (R2R) - Month-End/Year-End Financial Close
Submitted By	**Roll Number**	**Course**	**Branch**
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SCOPE	SAP S/4HANA 2023 | FI-GL · FI-AR · FI-AP · FI-AA · CO-CCA · CO-PCA | BTP: CAP · Integration Suite · HANA Cloud · Workflow Management · Build Apps

R2R	Record-to-Report is the end-to-end financial process that begins with recording business transactions and ends with delivering statutory financial statements. It is one of the most critical processes in any ERP-enabled organisation, directly impacting regulatory compliance, investor confidence, and management decision-making.

Problem Statement

Traditional SAP R2R financial close processes at mid-sized Indian companies like Vinayak Associate suffer from manual Excel-based task tracking, paper-driven journal entry approvals, delayed bank reconciliation, and a lack of real-time visibility for branch managers. This report documents the complete SAP FI/CO configuration and a five-application SAP BTP extension layer designed to compress the month-end close from 7+ days to a target of 4 calendar days - while ensuring full compliance with Ind AS, Companies Act 2013, Income Tax Act, and the GST Act.

Tech Stack

Attribute	Details
SAP S/4HANA 2023	Core ERP — Financial & Management Accounting
SAP FI (GL, AR, AP, AA)	Financial accounting, sub-ledgers, asset management
SAP CO (CCA, PCA, PA)	Cost centres, profit centres, profitability analysis
SAP CAP (Node.js)	Backend OData V4 API for Close Cockpit
SAP Fiori Elements	Frontend UI — Close Cockpit & Mobile App
SAP Integration Suite (CPI)	Bank statement automation & workflow triggers
SAP HANA Cloud	Analytics database & KPI calculation views
SAP Build Apps (PWA)	Mobile close status app for branch managers
SAP FCCX	Financial Closing Cockpit — task tracking & sign-off

1. Company Blueprint — Vinayak Associate

The company blueprint defines legal, organisational, and logistics entities mapping real-world business structures into SAP. For Vinayak Associate - a multi-branch trading and consulting firm - it is designed to support multi-ledger accounting, inter-branch cost allocation, and SAP BTP cloud extensions.

Parameter	Detail	SAP Object	Value
Company Name	Vinayak Associate	**Chart of Accounts**	VKCA 6-digit scheme
Company Code / CA	VK01 / VK01	**FY Variant**	V3 Apr–Mar + 4 Spl. Periods
Location	Lucknow, U.P. 226001	**Post./Field Status Var.**	VK01 / VKFS
Branches	Kanpur │ Varanasi │ Delhi NCR	**Leading / Parallel Ledger**	0L Ind AS │ 2L —IFRS
Industry	Trading, Consulting & Fin. Svcs	**Plants**	VKP1 Lucknow │ VKP2 Kanpur
Currency / FY	INR │ April 1 – March 31	**Sales / Purch. Org.**	VKSO │ VKPO (Centralised)
Legal Form / CIN	Pvt. Ltd. │ U74999UP2016PTC083421	**Credit Control Area**	VK01 | Vinayak Credit Control

2. Chart of Accounts & GL Configuration

CoA VKCA follows Ind AS Schedule III (Companies Act 2013) with a 6-digit numbering convention. GL accounts are classified by type (Asset, Liability, Revenue, Expense) for automatic financial statement generation. Key T-Codes: OB13 → OB62 → OB29 → OB37 → OBBO → OB52. Parallel Ledgers: 0L (Ind AS / INR) as leading ledger; 2L (IFRS / INR+USD) as non-leading — with IFRS 16 lease, IAS 19 benefit provisions, and fair value adjustments posting selectively to Ledger 2L via T-Code FINSC_LEDGER / FINSC_LD_GRP.

Account Range	Account Class	SAP Type	Key Examples
100000–149999	Fixed Assets	Asset (A)	Land, Buildings, IT Equipment, Vehicles
150000–199999	Current Assets	Asset (A)	Cash, Bank, AR, Inventory, Prepaid Expenses
200000–249999	Capital & Reserves	Liability (P)	Share Capital, Retained Earnings, General Reserve
250000–299999	Non-Current Liabilities	Liability (P)	Long-term Loans, Deferred Tax Liability
300000–349999	Current Liabilities	Liability (P)	Trade Creditors, GST Payable, Provisions
400000–499999	Revenue Accounts	Revenue (X)	Trading Income, Consulting Fees, Other Income
500000–699999	Operating Expenses	Expense (K)	COGS, Salaries, Rent, Depreciation, Admin Costs
700000–799999	Financial Items	Both	Interest Exp/Income, FX Gain/Loss, Bank Charges

3. Sub-Ledger & Controlling Configuration

Sub-ledgers (AR, AP, AA) always reconcile in real-time with corresponding GL reconciliation accounts. The CO module provides cost centre tracking, profit centre P&L, overhead allocation, and profitability analysis.

3.1 Accounts Receivable (FI-AR)

Account group CUST, number range 200000–299999, reconciliation GL 150010. Payment terms VK30/VK45 (Net 30/45 days). Dunning procedure VKDP: 4 levels (7/14/21/30 days) via F150. FX revaluation of open USD items at month-end via F.05.

3.2 Accounts Payable (FI-AP)

Account group VEND, reconciliation GL 300010. Auto Payment Program F110 configured with House Bank VKBI (SBI Lucknow, A/c: 987654321001), supporting NEFT/RTGS/Cheque. GR/IR clearing via MR11 monthly.

3.3 Fixed Asset Accounting (FI-AA)

Class	Description	Depreciation	GL Acct	Notes
Class 1000	Land	Not Depreciated	120010	Freehold - not subject to wear
Class 2000	Buildings	SLM 5% / 20 yrs	120020	Office premises Lucknow & Kanpur
Class 3000	Office Equipment	SLM 33.33% / 3 yrs	120030	Printers, scanners, furniture
Class 4000	Vehicles	WDV 25.89%	120040	Company cars & delivery vehicles
Class 5000	Computers & IT	SLM 33.33% / 3 yrs	120050	Laptops, servers, network equipment
Class 9000	Assets Under Construction	No Depreciation	120090	Capitalised once project is complete

Three depreciation areas: Area 01 (Ind AS), Area 15 (IFRS), Area 20 (Income Tax Act). GL assignment via AO90. Monthly depreciation run: AFAB. AUC capitalisation: AIAB/AIBU.

3.4 Controlling (CO) Module

Cost Centre Hierarchy VNKAS_HIER: VNK-CORP (Finance/HR/IT), VNK-TRADE (Lucknow/Kanpur), VNK-CONSULT, VNK-FINSERV, VNK-ADMIN. Profit Centres: PC-TRADE │ PC-CONSULT │ PC-FINSERV. Key T-Codes: OKKP/OX19 → KS01/KSH1 → KCH1/KE51 → KA01/KA06 → OKB9 → OKP1. Default PC substitution on B/S accounts via T-Code 3KEH.

4. Month-End Closing - Step-by-Step Process

The Month-End Close (MEC) targets a 4-calendar-day cycle managed via SAP Financial Closing Cockpit (FCCX), assigning tasks to responsible users with real-time status tracking and a full digital audit trail.

TIMELINE: Day –1: Pre-close prep → Day 0: Month-end → Day +1: FI Sub-ledger close → Day +2: CO close → Day +3: Reporting & sign-off

#	Activity	T-Code	Mod.	Responsible
Phase 1 — Pre-Close Preparation (Day –1)				
P1	Review & Clear GR/IR Clearing Account	MR11	FI-MM	AP & Procurement Team
P2	Automatic Clearing of Open Items (F.13) + Post Accruals (FBS1)	F.13/FBS1	FI-GL	Finance Team

P3	Verify Zero Intercompany Balances + Confirm No Parked Docs	FAGLB03/FBV0	FI-GL	Finance Controller
Phase 2 - FI Sub-Ledger Closing (Day 0 – Day +1)				
F1	FX Revaluation — Open AR Items (F.05) & AP Items (F.07)	F.05/F.07	FI-GL	Finance Controller
F2	Post Provision Doubtful Debts (F-02) + Dunning Run (F150)	F-02/F150	FI-AR	AR Team
F3	Auto Payment Run NEFT/RTGS (F110) + Bank Statement (FF_5/FEBP)	F110/FF_5	FI-AP/Bank	AP & Treasury
F4	Monthly Depreciation Posting + Sub-Ledger Reconciliation to GL	AFAB/FAGLB03	FI-AA/GL	Asset Accountant
Phase 3 - CO Module Closing (Day +1 – Day +2)				
C1	Calculate WIP (KKAX) + Cost Variances on Orders (KKS1)	KKAX/KKS1	CO-PC	Costing Team
C2	Settle Internal Orders to Cost Centres (KO88)	KO88	CO	Finance Controller
C3	Run Overhead Assessment (KSU5) + Distribution (KSV5) Cycles	KSU5/KSV5	CO-CCA	Mgmt. Accounting
C4	Transfer Actual Costs to CO-PA (KE5T) + Lock CA Period (OKP1)	KE5T/OKP1	CO-PA/CO	Mgmt. Accounting
Phase 4 - GL Closing & Period Lock (Day +2 – Day +3)				
G1	Post Remaining Manual Journal Entries (FB50) + Review Trial Balance (F.01)	FB50/F.01	FI-GL	Finance Controller
G2	Generate B/S & P&L (S_ALR_87012284) + Close Posting Period (OB52)	S_ALR/OB52	FI-GL	CFO / Finance Head

5. Year-End Closing - Step-by-Step Process

The Year-End Close (YEC) occurs at 31 March, encompassing all MEC Period 12 activities plus statutory, regulatory, and accounting steps unique to year-end. Compliance with Companies Act 2013, Income Tax Act, and Ind AS is mandatory.

KEY DATES: FY End: 31 March | Carryforward Target: 15 April | Board Approval: 30 June | Statutory Audit Sign-Off: 30 September (Companies Act 2013)

#	Activity	T-Code	Mod.	Responsible
Y01	Complete Final Month-End Close - Period 12 (March)	Multiple	FI/CO	Finance Controller
Y02	FX Revaluation - All Open Items at 31 March Rate (FAGL_FC_VAL)	FAGL_FC_VAL	FI-GL	Finance Controller
Y03	Post Tax Provision & Deferred Tax (FB50) + Employee Benefits (Gratuity/Leave)	FB50	FI-GL	Tax / HR Finance
Y04	Physical Inventory - Freeze (MI01/MI04) → Post Differences (MI07) → NRV (FB50)	MI01–MI07	MM/FI	Stores & Finance

Y05	Final Settlement Internal Orders (KO88) + CO Assessment/Distribution Cycles	KO88/KSU5	CO	Mgmt. Accounting
Y06	Profit Centre Year-End Closing Entry - Retained Earnings (9KE0)	9KE0	CO-PCA	Mgmt. Accounting
Y07	Open Special Periods 13-16 → Post Audit/Tax/IFRS Adjustments → Close Periods	OB52/FB50	FI-GL	Finance Controller
Y08	New GL Balance Carryforward (FAGLGVTR) + AR/AP Carryforward (F.07)	FAGLGVTR	FI-GL	Finance Controller
Y09	AA Year-End Close (AJAB) + Open New FY (AJRW)	AJAB/AJRW	FI-AA	Asset Accountant
Y10	Generate Final Audited Financial Statements + Lock Prior FY + Archive (SARA/DART)	F.01/OB52	FI-GL	CFO / Compliance

Period	Purpose	Detail & Statutory Basis
Period 13	Statutory Audit Adjustments	External audit corrections, prior-period rectifications per Ind AS 8
Period 14	Income Tax Audit (Sec. 44AB)	Deferred tax revisions, transfer pricing adjustments
Period 15	Group Consolidation & IFRS	Intercompany eliminations, IFRS vs. Ind AS reconciliation
Period 16	Board / Management Approvals	Exceptional items, Board-approved corrections & reclassifications

T-Code FAGLGVTR carries all B/S balances to Period 0 of the new FY; P&L accounts reset to zero; net profit/loss transfers to Retained Earnings GL 200050. Asset Accounting: AJAB (close) + AJRW (open new year).

6. SAP BTP Developer Extension — Financial Close Suite

The BTP Extension Suite builds cloud-native applications on SAP BTP - without modifying the S/4HANA core (Clean Core principle) - to automate and enhance the R2R close process. BTP Stack: SAP CAP (Node.js) │ Integration Suite (CPI) │ HANA Cloud │ Build Apps │ Workflow Management │ XSUAA / SAP IAS │ Alert Notification.

Application	Description & Key Features
App 1 — Financial Close Cockpit (Fiori + CAP)	Custom Fiori Elements (List Report + Object Page) on BTP HTML5 Repository. CAP Node.js OData V4 backend (CloseTaskService) reading from FCCX + HANA Cloud (CLOSE_TASKS, TASK_LOGS). XSUAA roles: CLOSE_VIEWER / CLOSE_EXECUTOR / CLOSE_APPROVER. Features: RAG status, S/4HANA deep-links, Gantt timeline, CFO PDF export.
App 2 — JE Approval Workflow (BTP Workflow Mgmt.)	All manual JEs > INR 1,00,000 require dual-control digital approval. Integration Suite iFlow detects parked document (FBV1) via RFC → starts Workflow instance → assigns User Task in Fiori My Inbox → upon approval posts document via OData API. Full approval trail retained as internal audit evidence.
App 3 — Bank Statement Integration	Timer-triggered iFlow at 08:00 IST connects to SBI Corporate API (OAuth 2.0 REST Adapter), retrieves bank statement, transforms SBI JSON → MT940 via Groovy script, sends to S/4HANA via IDoc Adapter (FINSTA01) — fully

(Integration Suite)	replacing manual FF_5 upload. Alert Notification on failure.
App 4 — Close Status Mobile App (SAP Build Apps)	PWA deployed on BTP HTML5 Repository for branch managers (Kanpur/Varanasi) without SAP GUI access. Authenticated via SAP IAS (SSO). Consumes same CAP OData V4 API as Close Cockpit. Features: period selector, RAG dashboard, overdue push notifications.
App 5 — Close Analytics Dashboard (HANA Cloud)	Delta-load iFlows replicate close data from S/4HANA to HANA Cloud. HANA Calculation View CV_CLOSE_KPI_DASHBOARD (params: FISCAL_YEAR, PERIOD, COMPANY_CODE) aggregates six KPIs consumed by a Fiori analytical card on the CFO dashboard.

KPI	Definition & Measurement	Target
Days-to-Close	Calendar days from period-end to final CFO sign-off	≤ 4 days
Journal Entry Volume	Count of manual JEs per period (automation benchmark)	Reduce 20% YoY
Rework Rate	% of posted JEs subsequently reversed or corrected	< 2%
JE Approval SLA	Elapsed time from FBV1 parking to BTP Workflow approval	< 2 hours
Sub-Ledger Recon. Time	Sub-ledger close start to GL reconciliation sign-off	< 4 hours
Audit Findings	Statutory audit observations raised per financial year	Zero repeats

7. FI Integration, Reports, T-Code Reference & Conclusion

7.1 FI-MM & FI-SD Integration

Every MM goods movement and SD billing document auto-generates FI postings via SAP Automatic Account Determination. FI-MM: MIGO (Dr Inventory/Cr GR-IR, OBYC BSX+WRX), MIRO (clears GR-IR/Cr Vendor), MB1A (Dr COGS, OBYC GBB-VBR), MI07 (Inv. Diff., OBYC INV), MR11 (GR/IR month-end clearing). FI-SD: VF01 (Dr Customer/Cr Revenue, VKOA ERE), VL02N (Dr COGS/Cr FG, VKOA EIN), F-29 (Down payment Spl. GL indicator A), F-32 (Cash discount auto-cleared).

7.2 Key Financial Reports

T-Code	Report & Purpose	Frequency
F.01 / S_ALR_87012284	Balance Sheet & P&L (Ind AS Schedule III)	Monthly / Annual
S_ALR_87012271	Cash Flow Statement - Indirect Method (Ind AS 7)	Annual
S_ALR_87012178/083	AR Aging / AP Aging (0–30, 31–60, 61–90, 90+ days)	Monthly
S_ALR_87011963	Fixed Asset Register - Cost, Accum. Dep., NBV, WDV	Monthly
S_ALR_87013611	Cost Centre Actual vs. Plan - Variance Analysis	Monthly
S_ALR_87013336	Profit Centre Divisional P&L - Trade/Consult/FS	Monthly
KE30 / FAGLB03	CO-PA Contribution Margin │ GL Account Balances (New GL)	Monthly

7.3 Master T-Code Quick Reference

T-Code	Description	T-Code	Description

OB52	Open / Close Posting Periods	**AFAB**	Periodic Depreciation Run
FAGLGVTR	New GL Balance Carryforward	**AJAB / AJRW**	AA Year-End Close & New Year
F.05 / F.07	FX Revaluation AR / AP	**F110 / F150**	Auto Payment Run / Dunning
FCCX	Financial Closing Cockpit	**FF_5 / FEBP**	Bank Statement Upload & Post
KSU5 / KSV5	CO Assessment & Distribution	**KO88**	Settle Internal Orders
MR11 / MI01-07	GR/IR Clearing / Physical Inv.	**SARA / DART**	Archiving / Tax Audit Data
FB50/FBS1/FBV1	GL JE, Accrual & Parked Docs	**FBL3N/FBL5N/FBL1N**	GL / Customer / Vendor Items

Conclusion

This report has presented a comprehensive, end-to-end documentation of the Record-to-Report (R2R) process for Vinayak Associate, Lucknow. The document covered the full SAP FI/CO configuration stack - Chart of Accounts VKCA, parallel ledger setup (Ind AS + IFRS), AR/AP/Asset Accounting sub-ledgers, Controlling module with cost and profit centres, a structured 4-day month-end close process across four phases, a 21-step year-end close with special period management, and a five-application SAP BTP Developer Extension Suite (Close Cockpit, JE Approval Workflow, Bank Statement Automation, Mobile Close App, and HANA Analytics Dashboard).

The SAP BTP layer applies the Clean Core principle - extending S/4HANA capabilities through cloud-native services without touching the core system - reducing the Days-to-Close target from 7+ days to 4 days, improving audit trail completeness, and providing real-time financial close visibility to branch managers and executive leadership alike. The entire solution ensures full compliance with Ind AS, the Companies Act 2013, the Income Tax Act, and the GST Act applicable to Indian companies.